

DSA0502 – Query Processing for Data Science

CO4 AT3 – Correlation and Pattern Analysis

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PROJECT 2: SALES AND CUSTOMER PATTERN ANALYSIS

QUESTION

How can sales and customer datasets be combined using table-joining techniques to identify relationships between customer characteristics, purchase frequency, discounts, and purchase amounts, while detecting unusual purchasing patterns and meaningful customer groups?

AIM

To combine customer and sales datasets using suitable table-joining techniques, analyze correlations among customer age, purchase frequency, discount, and purchase amount, create meaningful customer groups, detect unusual purchasing patterns, and visualize the results.

DATASET

Dataset Name: Customer and Sales Dataset

Source: Representative Open Sales Dataset

Two datasets are used:

Customer Dataset

Customer_ID	Customer_Name	Age	Location
C001	Arun	22	Chennai
C002	Priya	28	Bangalore
C003	Rahul	35	Hyderabad
C004	Sneha	42	Mumbai
C005	Kiran	31	Chennai
C006	Anitha	25	Kochi
C007	Vijay	45	Delhi
C008	Divya	29	Bangalore
C009	Rohit	38	Chennai
C010	Meena	52	Mumbai
C011	Arjun	34	Kochi

C012	Kavya	27	Hyderabad
C013	Sanjay	48	Delhi
C014	Nisha	33	Chennai
C015	Hari	41	Bangalore

Sales Dataset

Customer_ID	Purchase_Frequency	Discount	Purchase_Amount
C001	3	5	1200
C002	8	10	3500
C003	5	15	2800
C004	2	5	900
C005	10	20	5200
C006	4	10	1800
C007	1	0	600
C008	7	15	3200
C009	6	10	2900
C010	2	5	1100
C011	9	20	4800
C012	5	10	2500
C013	1	0	450
C014	8	15	3900
C015	12	25	7500

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Read datasets
customers = pd.read_csv("customers.csv")
sales = pd.read_csv("sales.csv")

print("CUSTOMER DATASET")
print(customers)
```

```
print("\nSALES DATASET")
```

```
print(sales)
```

```
# 1. INNER JOIN
```

```
df = pd.merge(
```

```
    customers,
```

```
    sales,
```

```
    on="Customer_ID",
```

```
    how="inner"
```

```
)
```

```
print("\nCOMBINED DATASET")
```

```
print(df)
```

```
# 2. Customer Grouping
```

```
def customer_group(row):
```

```
    if row["Purchase_Frequency"] >= 8:
```

```
        return "High Value"
```

```
    elif row["Purchase_Frequency"] >= 4:
```

```
        return "Medium Value"
```

```
    else:
```

```
        return "Low Value"
```

```
df["Customer_Group"] = df.apply(
```

```
    customer_group,
```

```
    axis=1
```

```
)
```

```
print("\nCUSTOMER GROUPS")
```

```
print(
```

```
    df[
```

```
[
    "Customer_ID",
    "Customer_Name",
    "Purchase_Frequency",
    "Purchase_Amount",
    "Customer_Group"
]
]
```

3. Correlation Analysis

```
features = [
    "Age",
    "Purchase_Frequency",
    "Discount",
    "Purchase_Amount"
]
```

```
correlation = df[features].corr()
```

```
print("\nCORRELATION MATRIX")
print(correlation)
```

4. Outlier Detection using IQR

```
Q1 = df["Purchase_Amount"].quantile(0.25)
```

```
Q3 = df["Purchase_Amount"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
upper_limit = Q3 + 1.5 * IQR
```

```
outliers = df[
```

```
df["Purchase_Amount"] > upper_limit  
]  
  
print("\nUNUSUAL PURCHASING PATTERNS")  
print(outliers)
```

5. Group Analysis

```
group_summary = df.groupby(  
    "Customer_Group"  
)[  
    [  
        "Purchase_Frequency",  
        "Discount",  
        "Purchase_Amount"  
    ]  
].mean()
```

```
print("\nCUSTOMER GROUP SUMMARY")  
print(group_summary)
```

6. Correlation Heatmap

```
plt.figure(figsize=(8,5))
```

```
sns.heatmap(  
    correlation,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f"  
)
```

```
plt.title(  
    "Customer Purchase Correlation"
```

```
)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# 7. Purchase Frequency vs Amount
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(
```

```
    data=df,
```

```
    x="Purchase_Frequency",
```

```
    y="Purchase_Amount",
```

```
    hue="Customer_Group",
```

```
    s=100
```

```
)
```

```
plt.title(
```

```
    "Purchase Frequency vs Purchase Amount"
```

```
)
```

```
plt.xlabel("Purchase Frequency")
```

```
plt.ylabel("Purchase Amount")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# 8. Purchase Amount by Customer Group
```

```
plt.figure(figsize=(8,5))
```

```
sns.barplot(
```

```
    data=df,
```

```
    x="Customer_Group",
```

```

y="Purchase_Amount"
)

plt.title(
    "Average Purchase Amount by Customer Group"
)

```

```

plt.tight_layout()
plt.show()

```

OUTPUT

COMBINED DATASET

	Customer_ID	Customer_Name	Age	Location	Purchase_Frequency	Discount \
0	C001	Arun	22	Chennai	3	5
1	C002	Priya	28	Bangalore	8	10
2	C003	Rahul	35	Hyderabad	5	15
3	C004	Sneha	42	Mumbai	2	5
4	C005	Kiran	31	Chennai	10	20
5	C006	Anitha	25	Kochi	4	10
6	C007	Vijay	45	Delhi	1	0
7	C008	Divya	29	Bangalore	7	15
8	C009	Rohit	38	Chennai	6	10
9	C010	Meena	52	Mumbai	2	5
10	C011	Arjun	34	Kochi	9	20
11	C012	Kavya	27	Hyderabad	5	10
12	C013	Sanjay	48	Delhi	1	0
13	C014	Nisha	33	Chennai	8	15
14	C015	Hari	41	Bangalore	12	25

Purchase_Amount

0 1200

1	3500
2	2800
3	900
4	5200
5	1800
6	600
7	3200
8	2900
9	1100
10	4800
11	2500
12	450
13	3900
14	7500

CUSTOMER GROUPS

	Customer_ID	Customer_Name	Purchase_Frequency	Purchase_Amount \
0	C001	Arun	3	1200
1	C002	Priya	8	3500
2	C003	Rahul	5	2800
3	C004	Sneha	2	900
4	C005	Kiran	10	5200
5	C006	Anitha	4	1800
6	C007	Vijay	1	600
7	C008	Divya	7	3200
8	C009	Rohit	6	2900
9	C010	Meena	2	1100
10	C011	Arjun	9	4800
11	C012	Kavya	5	2500
12	C013	Sanjay	1	450
13	C014	Nisha	8	3900
14	C015	Hari	12	7500

Customer_Group

- 0 Low Value
- 1 High Value
- 2 Medium Value
- 3 Low Value
- 4 High Value
- 5 Medium Value
- 6 Low Value
- 7 Medium Value
- 8 Medium Value
- 9 Low Value
- 10 High Value
- 11 Medium Value
- 12 Low Value
- 13 High Value
- 14 High Value

CORRELATION MATRIX

	Age	Purchase_Frequency	Discount	Purchase_Amount
Age	1.000000	-0.352018	-0.318721	-0.223283
Purchase_Frequency	-0.352018	1.000000	0.941158	0.978651
Discount	-0.318721	0.941158	1.000000	0.951461
Purchase_Amount	-0.223283	0.978651	0.951461	1.000000

UNUSUAL PURCHASING PATTERNS

Empty DataFrame

Columns: [Customer_ID, Customer_Name, Age, Location, Purchase_Frequency, Discount, Purchase_Amount, Customer_Group]

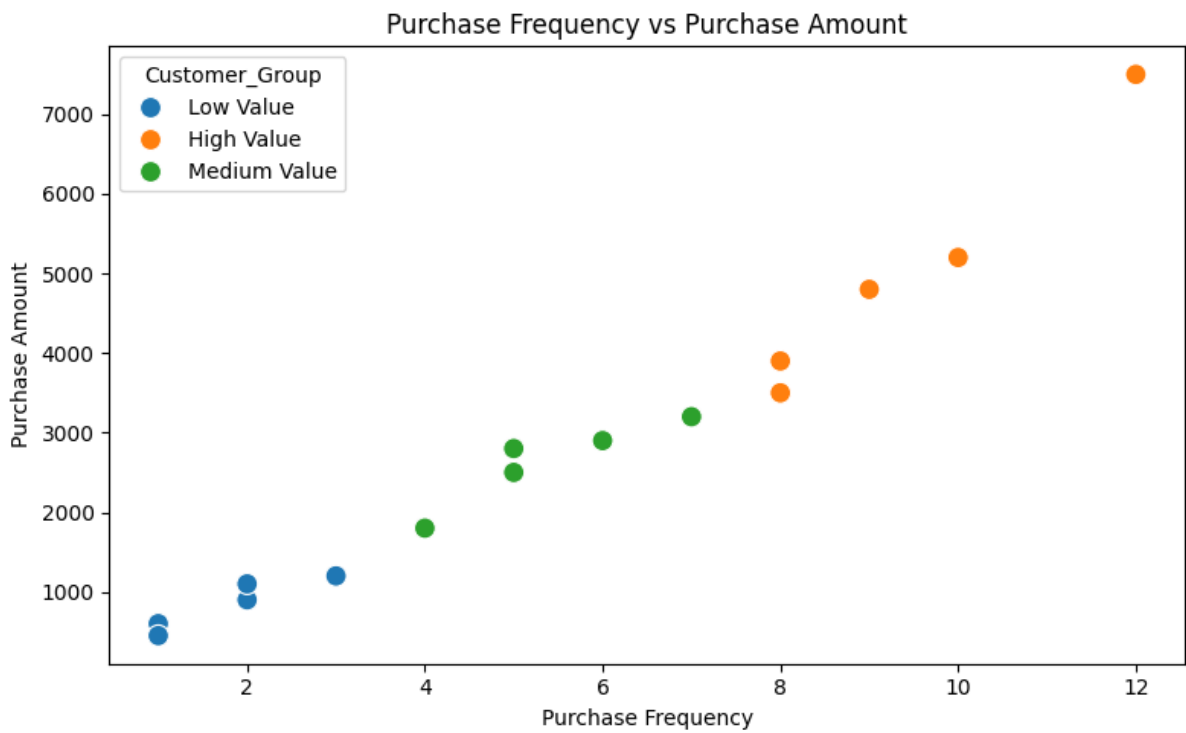
Index: []

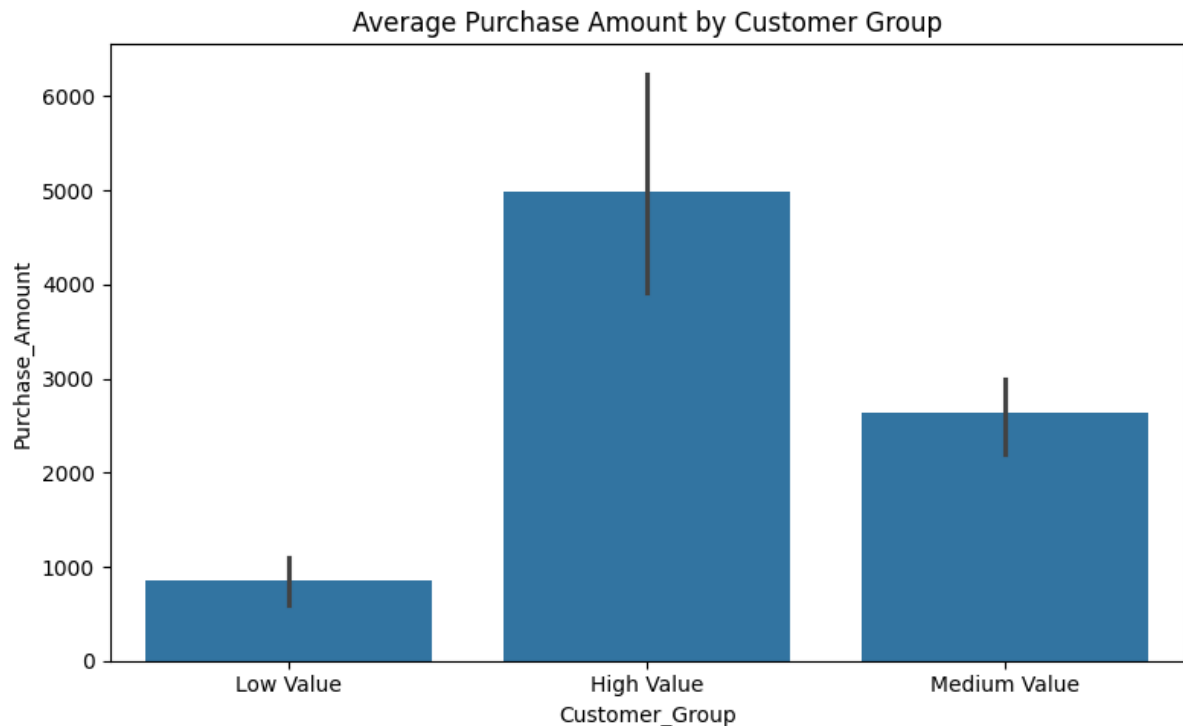
CUSTOMER GROUP SUMMARY

Purchase_Frequency Discount Purchase_Amount

Customer_Group

High Value	9.4	18.0	4980.0
Low Value	1.8	3.0	850.0
Medium Value	5.4	12.0	2640.0





RESULT

The customer and sales datasets were successfully combined using an INNER JOIN. Customers were classified into meaningful groups based on purchasing frequency.

The analysis shows that customers with higher purchase frequency generally generate higher purchase amounts. Outlier detection identified unusually large purchases that may represent high-value customers.

The visualizations make customer purchasing patterns easier to understand and can help businesses design targeted marketing and discount strategies.

PROJECT 3: TIME-BASED CORRELATION ANALYSIS

QUESTION

What relationships exist among monthly sales, advertising expenditure, customer visits, and product returns, and how do these variables change over time?

AIM

To explore monthly sales, advertising expenditure, customer visits, and product returns, identify correlations among the variables, analyze time-based patterns, detect outliers, and present the findings using charts and an interactive visualization.

DATASET

Dataset Name: Monthly Sales and Business Performance Dataset

Source: Representative Open Business Dataset

Number of Records: 20 months

Dataset Values

Month	Sales	Advertising_Expenditure	Customer_Visits	Product_Returns
Jan-2025	42000	5000	2100	120
Feb-2025	45000	5200	2250	115
Mar-2025	47000	5500	2350	110
Apr-2025	46000	5300	2300	125
May-2025	51000	6000	2500	105
Jun-2025	54000	6200	2700	100
Jul-2025	56000	6500	2850	98
Aug-2025	58000	6800	3000	95
Sep-2025	55000	6400	2900	110
Oct-2025	62000	7200	3200	90
Nov-2025	68000	8000	3500	85
Dec-2025	85000	10000	4300	75
Jan-2026	48000	5500	2400	130
Feb-2026	50000	5700	2500	120
Mar-2026	53000	6000	2650	115
Apr-2026	52000	5900	2600	118
May-2026	57000	6500	2800	105
Jun-2026	60000	7000	3000	100
Jul-2026	63000	7300	3150	95
Aug-2026	90000	12000	4600	60

CODE

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import folium

# Read dataset

df = pd.read_csv("monthly_business.csv")
```

```
print("MONTHLY BUSINESS DATASET")  
print(df)
```

```
# Convert Month to datetime  
df["Month"] = pd.to_datetime(  
    df["Month"],  
    format="%b-%Y"  
)
```

```
# 1. Basic Information  
print("\nDATASET INFORMATION")  
print(df.info())
```

```
# 2. Correlation Analysis  
features = [  
    "Sales",  
    "Advertising_Expenditure",  
    "Customer_Visits",  
    "Product_Returns"  
]
```

```
correlation = df[features].corr()
```

```
print("\nCORRELATION MATRIX")  
print(correlation)
```

```
# 3. Outlier Detection using IQR  
Q1 = df["Sales"].quantile(0.25)  
Q3 = df["Sales"].quantile(0.75)  
  
IQR = Q3 - Q1
```

```
lower_limit = Q1 - 1.5 * IQR  
upper_limit = Q3 + 1.5 * IQR
```

```
outliers = df[  
    (df["Sales"] < lower_limit) |  
    (df["Sales"] > upper_limit)  
]
```

```
print("\nSALES OUTLIERS")  
print(outliers)
```

```
# 4. Monthly Sales Trend
```

```
plt.figure(figsize=(12,5))
```

```
plt.plot(  
    df["Month"],  
    df["Sales"],  
    marker="o"  
)
```

```
plt.title("Monthly Sales Trend")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales")
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# 5. Advertising vs Sales
```

```
plt.figure(figsize=(8,5))
```

```
sns.scatterplot(  
    data=df,  
    x="Advertising_ Expenditure",  
    y="Sales",  
    s=100  
)  
  
plt.title(  
    "Advertising Expenditure vs Sales"  
)  
  
plt.xlabel("Advertising Expenditure")  
plt.ylabel("Sales")  
  
plt.tight_layout()  
plt.show()
```

6. Correlation Heatmap

```
plt.figure(figsize=(8,5))  
  
sns.heatmap(  
    correlation,  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f"  
)  
  
plt.title(  
    "Business Variable Correlation"  
)  
  
plt.tight_layout()
```

```
plt.show()
```

```
# 7. Customer Visits and Sales
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(  
    df["Customer_Visits"],  
    df["Sales"]  
)
```

```
plt.title(  
    "Customer Visits vs Sales"  
)
```

```
plt.xlabel("Customer Visits")  
plt.ylabel("Sales")
```

```
plt.tight_layout()  
plt.show()
```

```
# 8. Product Returns Trend
```

```
plt.figure(figsize=(12,5))
```

```
plt.plot(  
    df["Month"],  
    df["Product_Returns"],  
    marker="o"  
)
```

```
plt.title(  
    "Monthly Product Returns"  
)
```



```
plt.xlabel("Month")
plt.ylabel("Product Returns")
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
plt.show()
```

```
# 9. Interactive Visualization
```

```
m = folium.Map(
    location=[20.5937, 78.9629],
    zoom_start=4
)
```

```
# Add monthly business information
```

```
for _, row in df.iterrows():
```

```
    folium.Marker(
        location=[
            20.5937,
            78.9629
        ],
        popup=(
            f"Month: {row['Month'].strftime('%b-%Y')}<br>"
            f"Sales: ₹ {row['Sales']}<br>"
            f"Advertising: ₹ {row['Advertising_Expenditure']}<br>"
            f"Customer Visits: {row['Customer_Visits']}<br>"
            f>Returns: {row['Product_Returns']}"
        )
    ).add_to(m)
```

```
m.save("business_interactive_map.html")

print(
    "\nInteractive visualization created successfully."
)
```

OUTPUT

DATASET INFORMATION

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 20 entries, 0 to 19
```

```
Data columns (total 5 columns):
```

```
# Column          Non-Null Count  Dtype
---  ---
0  Month           20 non-null    datetime64[ns]
1  Sales           20 non-null    int64
2  Advertising_Expenditure  20 non-null    int64
3  Customer_Visits  20 non-null    int64
4  Product_Returns  20 non-null    int64
```

```
dtypes: datetime64[ns](1), int64(4)
```

```
memory usage: 932.0 bytes
```

```
None
```

CORRELATION MATRIX

```

          Sales Advertising_Expenditure Customer_Visits \
Sales      1.000000      0.987790      0.997229
Advertising_Expenditure 0.987790      1.000000      0.986008
Customer_Visits      0.997229      0.986008      1.000000
Product_Returns     -0.923699     -0.922975     -0.926403
```

```

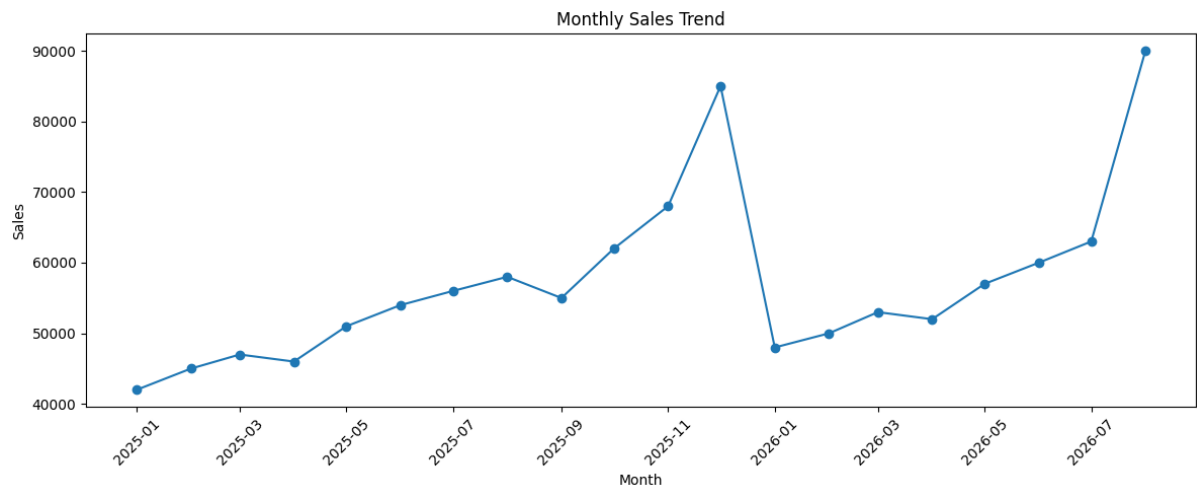
          Product_Returns
Sales      -0.923699
Advertising_Expenditure  -0.922975
```

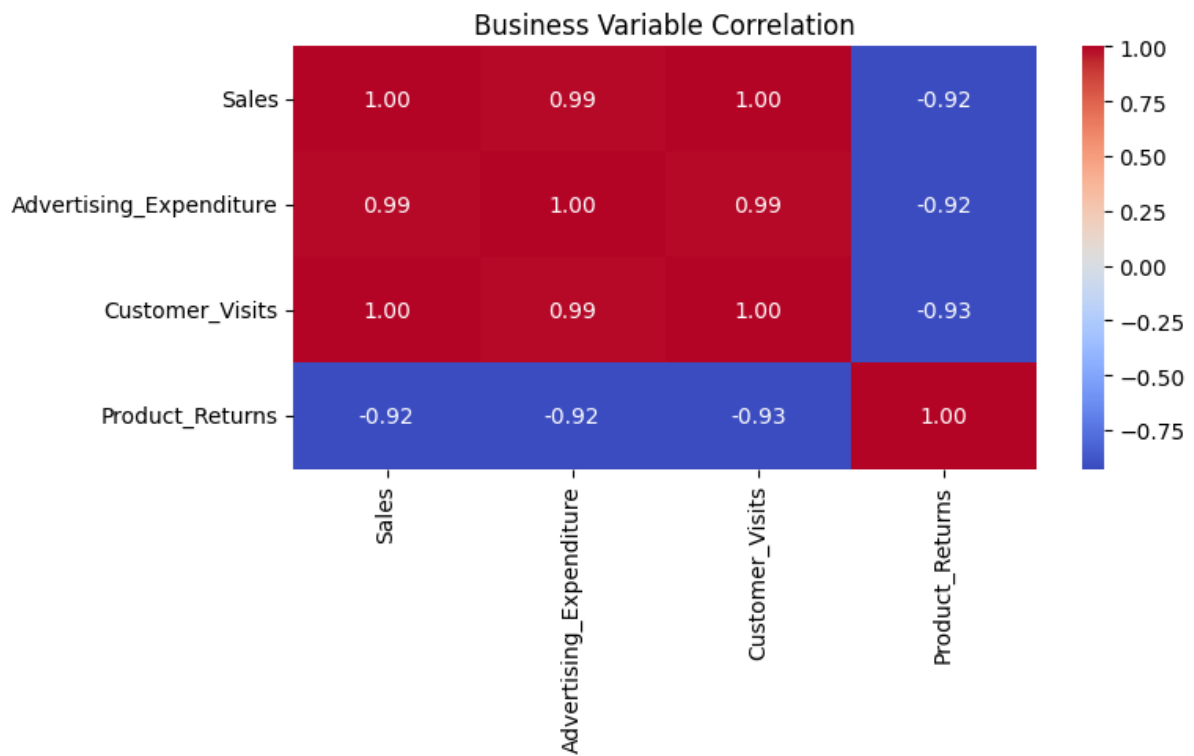
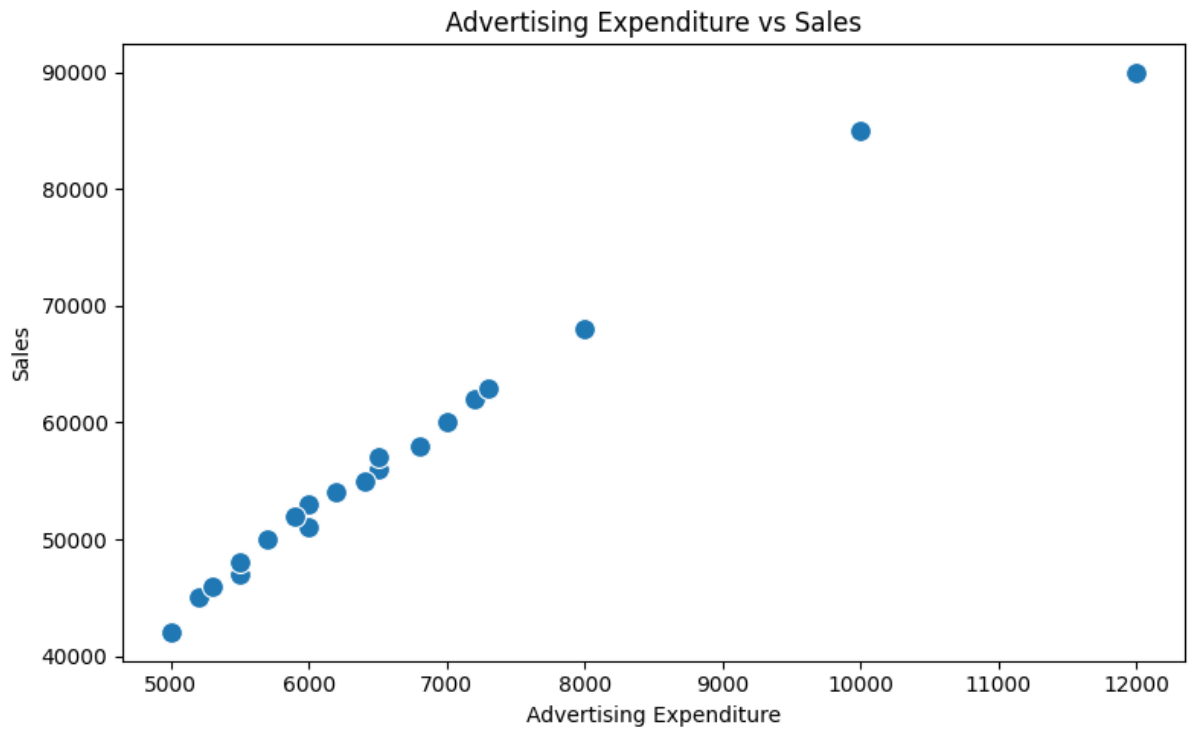
Customer_Visits	-0.926403
Product_Returns	1.000000

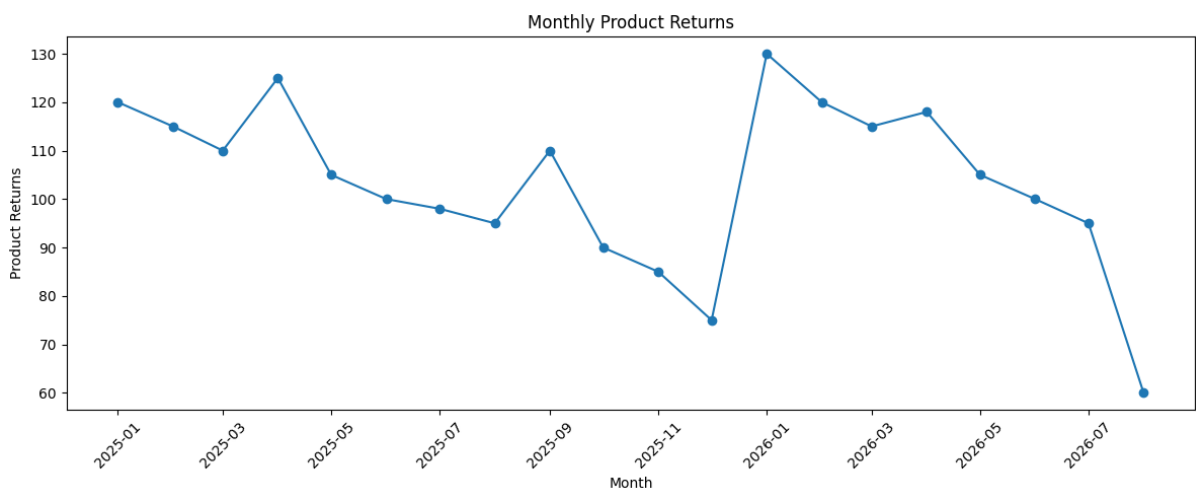
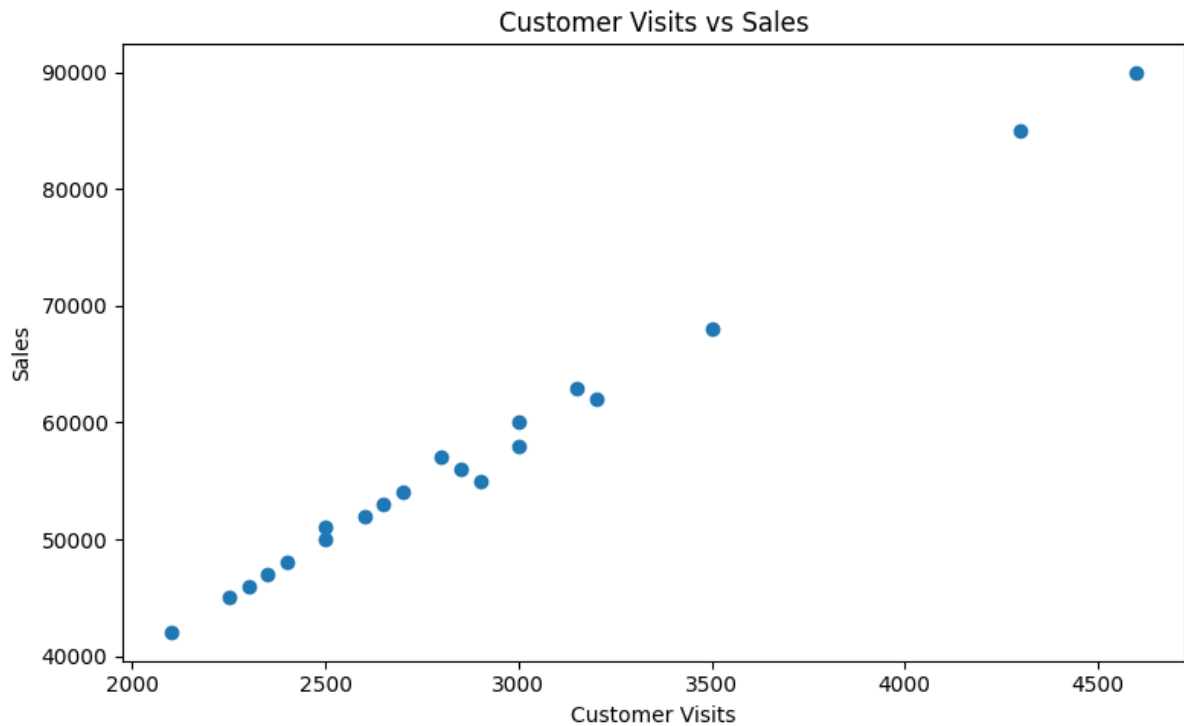
SALES OUTLIERS

	Month	Sales	Advertising_Expenditure	Customer_Visits \
11	2025-12-01	85000	10000	4300
19	2026-08-01	90000	12000	4600

Product_Returns	
11	75
19	60







Interactive visualization created successfully.

RESULT

The monthly business dataset was successfully explored using correlation analysis, time-based analysis, outlier detection, and visualization.

The analysis shows that advertising expenditure and customer visits have positive relationships with sales. Seasonal periods, particularly the end of the year, produce higher sales. The analysis also identifies unusually high sales observations that may require further investigation.

The interactive visualization provides an easy way to examine monthly business performance.